How the Brain Recovers: Adult Neurogenesis and the Science of Rehabilitation

From the birth of new neurons to appropriate connections and functional recovery—human tissue, rehabilitation trials, and the bottlenecks AI and quantum sensors aim to address.

RESEARCH REVIEW · OCTOBER 30, 2026 · NEUROBIOLOGY · BRAIN INJURY · REHABILITATION

Hyun-Jung Kim · AI-assisted · evidence cutoff 2026-10-30 · Production and verification



A small new neuron within a curved cell layer and the surrounding mature branches evoke local neuron formation and the challenge of joining an existing circuit. AI-generated conceptual illustration with a simplified hippocampal structure.

1. The birthdays of brain cells, recorded by nuclear tests

Atmospheric nuclear tests in the twentieth century left a radioactive carbon signature in the air. Plants absorbed that carbon, and the food chain carried it into human bodies. Decades later, researchers turned this unexpected chronology into a question about the brain: when were its cells born? Carbon incorporated into DNA when a cell divides can be compared with atmospheric concentrations at the time, allowing researchers to estimate the ages of cells living within a tissue. In 2013, Kirsty Spalding and colleagues used this method to report that a fraction of the neuronal population in the adult hippocampus undergoes lifelong turnover. [1]

Reading the birthdays of cells from postmortem tissue is remarkable. Yet the prospect of brain recovery immediately raises harder questions. Did a newly born cell survive? What kind of neuron did it become? Whose signals did it receive, and where did it send its own? What could a person do again because of that change?

Adult neuron formation, recently explored by New Scientist, invites precisely these hopes. Studies of the human hippocampus published in 2025 and 2026 added molecular and histological evidence to a long-running debate about whether the process persists. Assessing its potential now requires following the journey from a cell's birth to a working circuit and, ultimately, a change in everyday life. [2]

The possibility of new neurons forming in the adult hippocampus and the effectiveness of a treatment for restoring function after brain injury are distinct questions that can be studied together. Rehabilitation currently helps people use surviving circuits, learn strategies, and adapt tasks and environments. Regeneration research seeks biological ways to support or supplement those circuits. Understanding both makes the value of a new discovery more concrete.

2. An immature cell is not automatically a newly born cell

Adult neurogenesis is the process through which neural progenitor cells in a mature brain proliferate and their descendants differentiate into neurons, mature, and participate in existing circuits. A particularly well-studied site in humans is the dentate gyrus, a part of the hippocampus involved in learning and memory. Findings from this restricted region cannot be taken to mean that the cerebral cortex can rebuild its motor and language circuits in the same way.

In 1998, Peter Eriksson and colleagues examined the postmortem hippocampi of five adults who had received the labeling compound BrdU for cancer diagnosis during life. They reported cells that carried both a trace of DNA synthesis and neuronal markers. Repeatedly administering birth labels to people for research is difficult, so these rare tissue samples and radiocarbon dating offered different kinds of evidence. BrdU records DNA synthesis during the labeling period; radiocarbon dating examines the DNA ages of isolated cell populations through a model of turnover. [1,3]

The debate returned to what should count as evidence of a new neuron. The protein DCX is often used to identify immature neurons, but a DCX-positive cell cannot simply be assigned a recent adult birthday. Researchers must also consider whether it could have remained immature for a long time. Ki67 marks cells that are proliferating, but identifying their type and determining what they later become requires additional evidence. The number of immature cells observed at one moment reflects the rate of production, the length of time cells remain immature, their survival, and the conditions under which they can be detected.

In 2018, Shawn Sorrells and colleagues reported a steep decline in human hippocampal neurogenesis during infancy and childhood, with young neurons not detected in the adult dentate gyrus. That same year, Maura Boldrini and colleagues reported persistent immature neurons in the dentate gyrus of adults and older people. In 2019, Elena Moreno-Jiménez and colleagues experimentally showed that prolonged tissue fixation can make relevant markers harder to detect. [4,5,6]

Declaring a simple winner in this disagreement would erase the difficulty of studying the human brain. Health and disease before death, the exact tissue sampled, the postmortem interval, fixation, antibody specificity, and the criteria used to classify rare cells can all change what researchers see. Recent work matters because it goes beyond finding a stronger signal for a single marker: it asks whether several methods point to the same cellular identity.

3. What became visible in 2025 and 2026?

Single-nucleus analysis did not initially produce a consistent answer either. A cross-species study by Franjic and colleagues in 2022 found no clear neurogenic trajectory in the human dentate gyrus. Capturing sufficiently rare cells and deciding how to apply markers learned from other species became central challenges for subsequent studies. [7]

In a 2025 Science paper, Ionut Dumitru and colleagues combined single-nucleus RNA analysis, localization within tissue, and machine-learning classification to identify proliferating neural progenitor cells in the adult human hippocampus. Single-nucleus RNA analysis reads the RNA expressed by individual nuclei to infer cell types and states. To find rare progenitors, some samples underwent enrichment using a proliferation marker. [8]

The study analyzed material from six children and nineteen individuals aged 13-78. The combined childhood and adolescent-to-adult datasets contained 402,660 nuclei. Applying a classifier trained on the childhood data to 286,799 adolescent-to-adult nuclei identified 354 candidate neural progenitor cells. Finding a small population within such a large dataset was a major technical challenge. Those 354 candidates cannot, however, be converted into a daily rate of neuron birth. Nor can proportions from differently enriched samples be read as age-specific rates of neurogenesis. The researchers also identified a need to determine how much adult cell division and the slow maturation of cells generated much earlier each contribute. [8]

A 2026 Nature study by Ahmed Disouky and colleagues added measurements of chromatin accessibility to RNA analysis. Chromatin packages DNA with proteins; which regions are accessible provides clues about gene regulation. The researchers compared postmortem hippocampi from young adults, older people with age-appropriate cognition, SuperAgers with exceptional episodic memory, a preclinical pathology group, and people with Alzheimer's disease. The dataset comprised 355,997 **nuclei** from 38 donors. The number of people available for comparisons and the number of nuclei analyzed within their tissue must be kept separate. [9]

The study identified molecular states corresponding to neural stem cells, neuroblasts, and immature granule cells, with differences in gene regulation across cognitive and pathological groups. These findings provide specific targets for investigating how cellular states relate to healthy aging and disease. A postmortem, cross-sectional comparison cannot establish that SuperAgers' lifestyles increased neurogenesis and thereby preserved memory. [9]

Machine learning extends the microscope's reach in these studies. Researchers cannot manually classify the expression patterns of hundreds of thousands of nuclei, so computational comparisons with known cell states help locate rare candidates. Yet the species and ages used for training, marker genes, and reference datasets influence the result. RNA velocity uses newly transcribed and processed RNA to model the direction of changes in expression. Neither this inference nor a developmental ordering reconstructed from RNA states is a recording of one person's cells followed over time. It is a computational inference about molecular continuity that must be interpreted alongside lineage labeling capable of directly tracking cells and their descendants. [8,9]

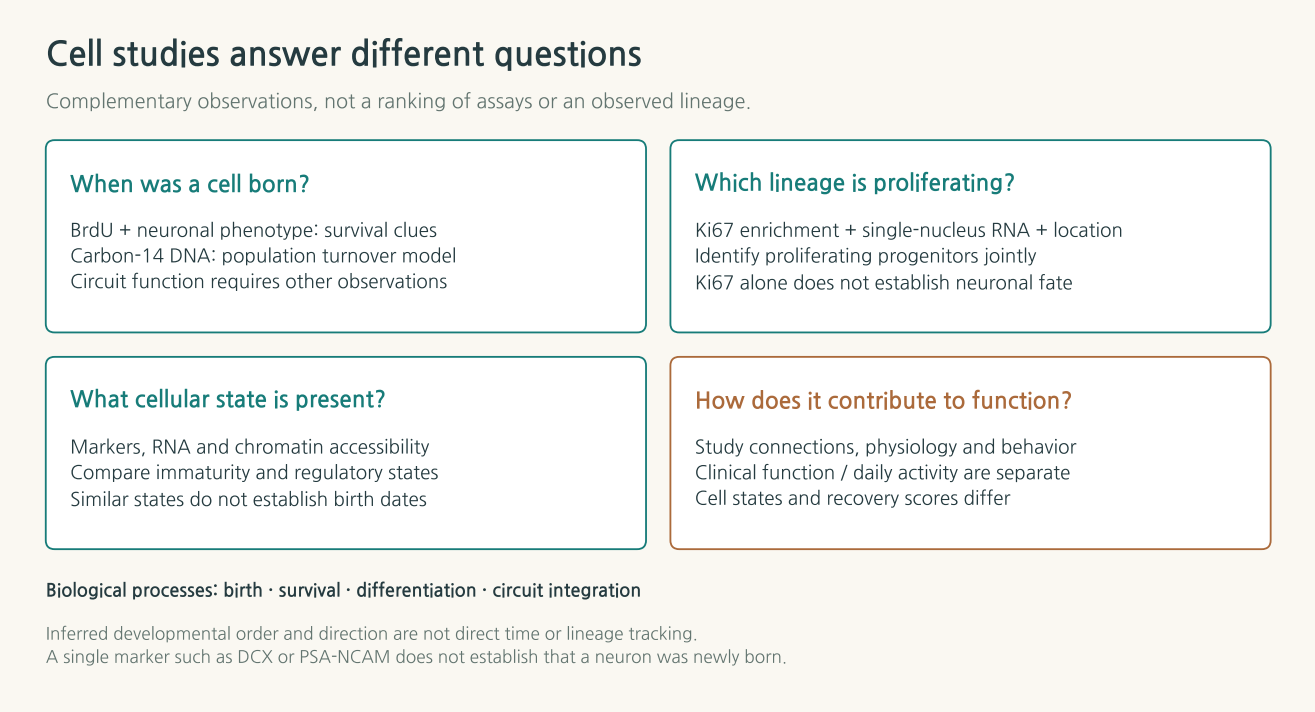


Figure 1. Evidence of cell birth, proliferation and differentiation, circuit integration, and functional change answers different questions. Explanatory diagram constructed for this review from human hippocampal tissue and molecular studies. The four areas represent different observational questions, not treatment success rates or a recorded lineage of one cell. [1,3,8,9]

Stronger evidence for cell birth and cellular states in Figure 1 makes questions about circuit function more tractable. Beyond identifying clues to birth and maturation, researchers must establish the connections cells make, the information they process, and whether their activity causes a functional change. The limited ability to repeatedly measure these relationships at cellular resolution in a living human brain remains a major obstacle to clinical translation.

4. What must a new cell do to become useful?

Granule cells in the dentate gyrus receive incoming signals and send their output to other hippocampal regions. One function discussed in relation to this system is pattern separation: distinguishing similar experiences. Remembering where a car was parked today without confusing it with yesterday's location suggests why this matters. A score on such a behavioral task cannot, however, be used to work backward to the number of new neurons a person has produced.

Circuit performance depends on more than making additional cells. A cell must reach an appropriate position and survive, receive inputs through its dendrites, and send outputs through its axon to suitable targets. The balance of excitatory and inhibitory inputs, and the timing of activity during learning, also matter. Brain injury can affect cell bodies, axons connecting distant regions, blood vessels, and the surrounding cellular environment. Increasing the supply of cells in one region and restoring a distributed circuit therefore pose different challenges.

Laura Villasana and colleagues illustrated this problem in a controlled cortical impact model of traumatic brain injury in mice. Hippocampal neurons born after injury showed electrical activity and synaptic inputs, but their locations and dendritic shapes differed from those in controls. The study demonstrated that these cells could join circuits while leaving open what their altered arrangement meant for information processing. It was not a trial of treatment efficacy in people. [10]

Even in animals with greater regenerative capacity, cell birth and functional recovery need not occur on the same schedule. A 2026 paper by Valerie Mariën and colleagues, whose public abstract reports a brain-injury model in the African turquoise killifish, combined birth labeling, electrophysiology, and learned avoidance behavior. The acquisition of mature cellular form and electrical properties, together with circuit integration, accompanied behavioral recovery; sufficient recovery took about 50 days. The brain region studied corresponds to the mammalian amygdala. That interval cannot be converted into a timetable for the human hippocampus or rehabilitation, but it sharpens the question of why function can recover later than the visible replacement of tissue. [11]

The next goal is to identify **the conditions under which the right cells function appropriately in the circuits that need them**. This makes the cellular environment, experience, and rehabilitation that uses those circuits relevant partners for regeneration research. Whether combining these approaches adds clinical benefit, and for whom, remains a question for trials.

5. Rehabilitation works with the brain that remains

Neuroplasticity is the brain's capacity to change its structure and function in response to experience and injury. The formation and integration of new neurons is one part of it. The plasticity relevant to rehabilitation also includes changes in existing synaptic strength and activity, learning strategies, and recruiting other circuits. These can be complemented by moving some memory demands outside the brain—for example, through smartphone reminders—or by changing the sequence of a task. [12]

Calling all these changes “brain regeneration” can obscure what has actually improved. Someone who keeps appointments more reliably may have gained memory capacity, learned a checking routine, or become better at using reminders. Each can help everyday life, but the training involved and the conditions needed to sustain its benefit differ. Neuron formation and functional benefit require their own appropriate measurements.

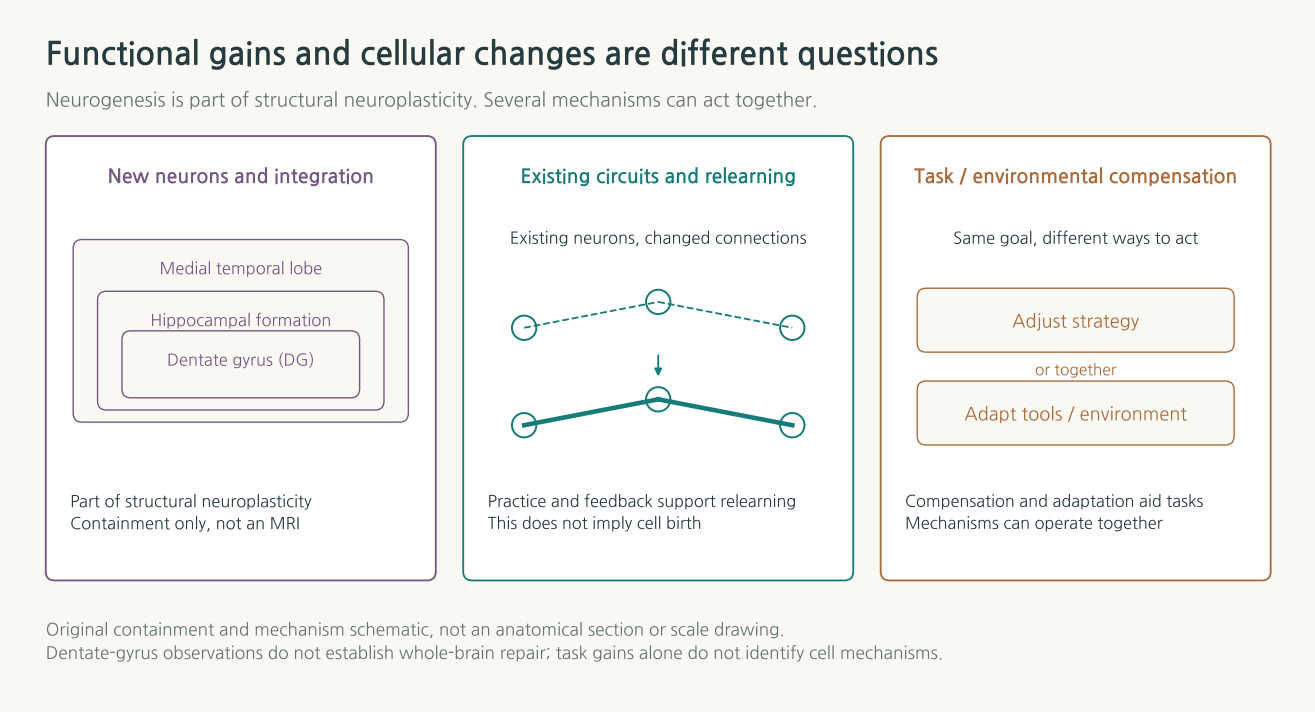


Figure 2. New cell formation and integration, adjustment and relearning within existing circuits, and changes to tasks and environments are potentially overlapping mechanisms and strategies for recovery. Neurogenesis is itself a form of neuroplasticity. Hippocampal locations and connections are conceptual. The diagram was constructed for this review from human tissue research and the INCOG memory rehabilitation guidelines. [8,9,12]

The INCOG 2.0 memory guidelines for cognitive rehabilitation after moderate-to-severe traumatic brain injury emphasize strategy training and external memory aids tailored to a person's actual tasks. Providing an electronic reminder is only the beginning: people need to learn when and how to check and use it. For severe memory impairment, learning methods that limit errors and support from the everyday environment also matter. Gains from computer practice must likewise be assessed for transfer to activities that the person needs to perform. [12]

Clinical findings direct attention to the change measured rather than the name of a program. The TBI-MEM randomized trial assigned 69 adults to memory strategy training using context and imagery, with 35 in the treatment group and 34 in the placebo-control group. Some story-memory and task outcomes improved, but one of the two coprimary outcomes—the learning slope for a word list—showed no significant difference. The usefulness of a particular strategy must therefore be distinguished from recovery of memory as a whole. [13]

The larger ReMemBrIn trial randomized 328 adults to test the addition of ten weeks of group memory rehabilitation to usual care. It did not demonstrate a between-group benefit on the primary measure of everyday forgetting at six months. The adjusted difference was −2.1 points, with a 95% confidence interval from −6.7 to 2.5, including zero, the value corresponding to no difference. Differences were reported in outcomes such as individual goal attainment. These findings encourage separate assessment of the participants, strategies, level of engagement, and transfer to everyday life, rather than a single verdict on all memory rehabilitation. [14]

The three branches in Figure 2 need not be ranked as competing treatments. Biological capacity, learned strategies, and the environment can all help someone perform a task more independently. Findings from adult trials also cannot simply be transferred to other ages or to injuries with different causes. A clinical team adjusts rehabilitation goals, intensity, and timing to the injury, fatigue, and accompanying symptoms.

6. How far has cell therapy already come?

It would also be inaccurate to say that every cell therapy for brain injury belongs to the distant future. On July 31, 2024, Japan granted conditional, time-limited approval to SB623, also called vandefitemcel and marketed as Akuugo, for improving chronic motor paralysis associated with traumatic brain injury. The approval carries a seven-year limit and postmarketing evaluation requirements. An official announcement reported the first postmarketing administration on October 2, 2026. [15,16]

The product involves intracranial implantation of modified, allogeneic bone marrow-derived mesenchymal cells. These are fundamentally different cells from hippocampal neural progenitors. Proposed actions include effects of secreted factors on host tissue repair, blood vessels, and inflammation; PMDA also concluded that the mechanism was not fully understood. Approval does not establish that the transplanted cells replaced neurons and rebuilt precise circuits in people. [15]

The supporting phase 2 STEMTRA trial enrolled adults with chronic motor deficits at least twelve months after injury. Of 63 randomized participants, 61 who underwent surgery—46 pooled across three cell-dose groups and 15 in the sham-surgery group—were included in the modified intention-to-treat analysis. At six months, the adjusted mean change in the Fugl-Meyer Motor Scale, a 0-100-point assessment of arm and leg motor function, with higher scores indicating better function, was +8.3 points in the cell group and +2.3 in controls. The reported between-group difference was 6.0 points, with a 95% confidence interval of 0.3-11.8. This is a clinical signal in a specific population with motor impairment; it does not imply an equivalent improvement in memory or overall independence in daily life. [17]

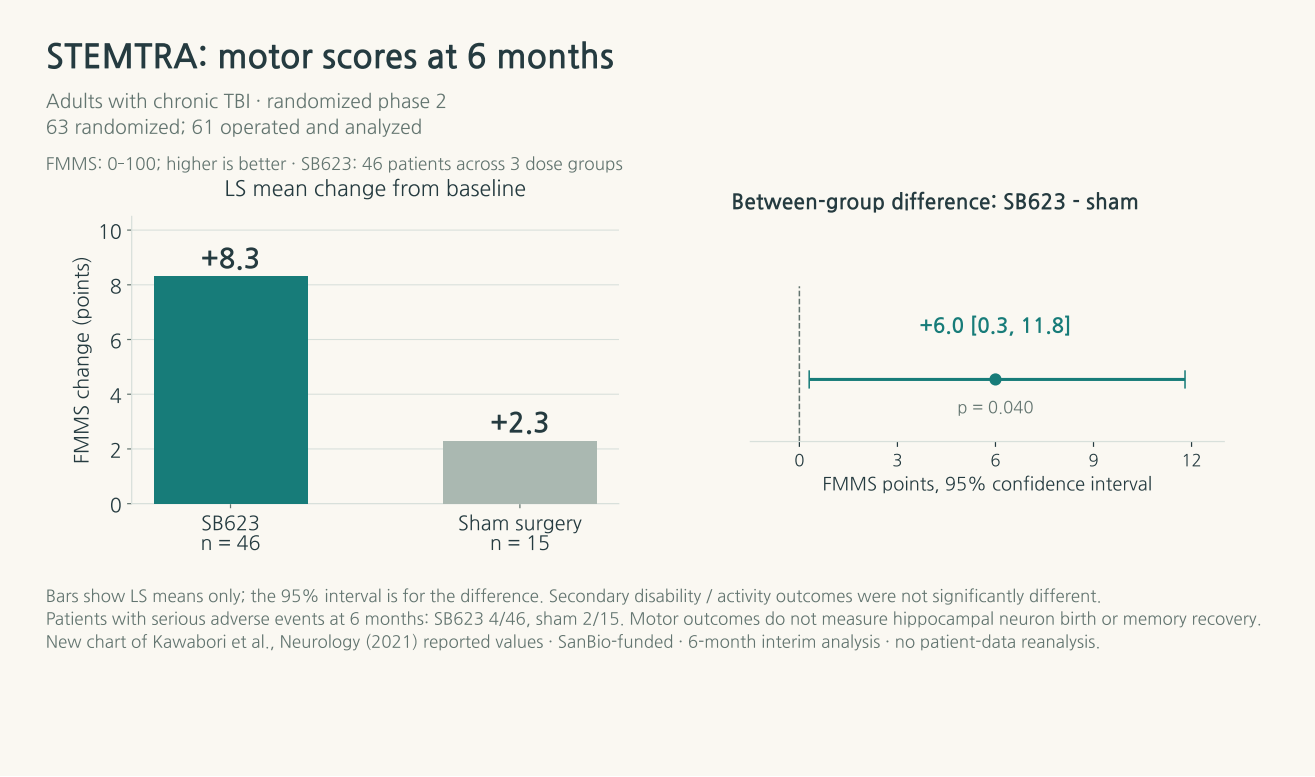


Figure 3. Reconstruction of the authors' reported phase 2 STEMTRA results. Left: least-squares mean changes in the Fugl-Meyer Motor Scale at six months. Right: the between-group difference and its 95% confidence interval. There were 63 randomized participants and 61 in the postsurgical analysis. The study did not measure memory or neuron birth, and between-group differences in secondary efficacy outcomes were not significant. [17]

Figure 3 requires attention to the width of the interval as well as the direction of the score. The estimated average effect is positive, but uncertainty remains in this small trial. Secondary disability and activity measures did not show significant between-group differences, and SanBio funded the study. The sham procedure did not reproduce the tissue penetration involved in cell injection. In the 24-week interim analysis, serious adverse events were reported in four of 46 cell-treated participants and two of 15 controls; surgery and imaging-detected hemorrhage also presented risks. Evaluation in larger populations and over longer periods is part of the rationale for conditional access. [15,17]

A 2024 report presented a post hoc analysis of the same trial at 48 weeks. Improvement from baseline persisted in the pooled cell group, but the between-group comparison—+7.5 points versus +4.1 in controls—was not statistically significant (p=0.20). Disability, activity and daily-function measures also showed no significant between-group differences. An exploratory result in the five-million-cell dose group suggests a direction for further research; it is neither a new independent trial nor a substitute for the pooled-group comparison. Through 48 weeks, serious adverse events had occurred in four of 46 cell-treated participants and three of 15 controls. [18]

Japan's current package insert describes usual use in adults and specifies, among other conditions, moderate-to-severe motor impairment that has become fixed and an injury that occurred at least six months earlier. STEMTRA recruited adults at least twelve months after injury, so its enrollment criteria must be distinguished from post-authorization conditions of use. [15,17] Scientifically, this case shows an approach directed at

regeneration being tested against human functional outcomes. It also demonstrates why conditional access to a particular product, the size and durability of clinical benefit, and an explanation invoking new neurons must each be evaluated separately.

7. Exercise and the promise of “growing brain cells”

Once adult neurogenesis enters public discussion, a natural question follows: what can increase it? The first step is to ask what a study actually measured. A change in hippocampal volume after exercise, or in blood levels of the neurotrophic factor BDNF, is not a direct count of neurons born in a living human brain. Blood vessels, cellular morphology, and several tissue components can contribute to an imaging measurement.

A frequently cited 2011 trial compared a year of aerobic exercise with stretching in 120 older adults and reported an approximately 2% increase in hippocampal volume in the exercise group. This was an important demonstration of human structural plasticity. Yet between-group differences in memory change were not significant, and the association between volume change and some memory outcomes did not establish an effect mediated by new neurons. The participants were not patients undergoing brain-injury rehabilitation. [19]

The NeuroFit randomized trial, published in September 2026, brought this translation problem into sharper focus. It assigned 52 healthy adults aged 45-65 to twelve weeks of exercise, with 27 in the exercise group and 25 controls. Alongside cognition and mood, the researchers assessed how participants' serum affected cultured human fetal-derived hippocampal progenitor cells. This approach examines how factors in human blood change the behavior of cells in culture. It does not measure cell birth in the living participants' hippocampi. [20]

There was no significant between-group change in the primary outcome, pattern separation. Depressive symptoms improved, while the culture experiment actually showed a decrease in the proportion of cells positive for the proliferation marker Ki67. The researchers discussed a possible regulatory effect related to long-term maintenance of the progenitor pool, but that interpretation requires further testing. The result cannot establish either an increase or a decrease in neurogenesis in the human brain. [20]

Exercise's various benefits, changes in a specific cognitive task, and changes in cultured cells exposed to serum can be investigated together, but they are different measurements. After brain injury, the type and intensity of exercise must also be tailored to the person's condition. A striking phrase drawn from general health research provides little basis for selecting a rehabilitation program or a self-directed treatment. Useful clinical information identifies what was done, in whom, and which functions changed by how much.

8. AI looks beyond cell counts to the conditions for useful connections

Observing the birth of a neuron leaves further questions: what kind of cell should it become, and where should it connect? AI has a concrete role in regeneration research when it compares gene expression and wiring patterns across many cells to narrow the conditions worth testing experimentally. This takes the analysis beyond classifying the rare progenitors discussed earlier, toward identifying biological rules for building connections.

In 2026, Zhifu Wang, Su-Chun Zhang and colleagues at Duke-NUS and Sanford Burnham Prebys transplanted human cortical progenitors into the ischemically injured motor cortex of mice. As the cells matured, they integrated into cortical and subcortical circuits, including the corticospinal tract. The researchers traced where the axons went and read single-nucleus RNA profiles, then used machine-learning regression to examine the relationship between gene expression and projection destinations. The subjects were human-derived cells transplanted from outside the brain; this finding concerns a different process from increasing endogenous neurogenesis in an adult hippocampus. [21]

The authors' publicly available work and CTIP2-knockdown data also document an experiment in which reducing the expression of this transcription factor changed grafted neurons' projection patterns. A transcription factor is a protein that regulates the activity of other genes. The result turns the goal of supplying neurons into a more specific question about jointly controlling a cell's identity and its output destinations. It does not represent an independent patient study validating a treatment recommended by AI. Integration of human cells within a mouse brain cannot establish a recovery benefit in people. [21,22]

Another approach targets cells responding to the injured environment. Margherita Zamboni and colleagues at Karolinska combined gene-expression and chromatin-accessibility measurements from injured mouse spinal cords, then trained deep-learning models to predict accessibility from DNA sequence. They identified enhancers —DNA sequences that regulate gene expression—whose activity combines information about injury and cell identity. These sequences allowed a gene-delivery vector to express its cargo selectively in reactive astrocytes, glial cells involved in the environment around neurons and the response to injury. [23]

Their selected IREN5 sequence also worked in a cortical stab-wound model. Among labeled cells, 98% were astrocytes, a measure of **targeting specificity** obtained in four mice. It does not mean that 98% of all astrocytes were labeled or that 98% of the injury recovered. The vector was given before injury; the experiments used a reporter rather than a therapeutic cargo and did not demonstrate axon regeneration or behavioral recovery. The advance was a molecular tool for concentrating gene expression in cells occupying a particular state after injury. Whether such targeting can deliver a useful treatment remains a separate experimental question. [23]

The importance of the destination also emerged from a 2023 mouse spinal-cord study by collaborators at EPFL/NeuroRestore, UCLA and Harvard. Machine-learning analysis of single-nucleus data helped prioritize a neuronal population associated with recovery. The team then combined genetic stimulation of growth with guidance signals. Growing axons across the lesion alone did not restore walking; guiding them back to their natural target region produced functional gains. This preclinical design included delivery of growth-promoting vectors before injury. It was neither a treatment applied to people with brain injury nor an experiment increasing the birth of neurons. In a different model—regrowth of existing neurons' axons—it showed why **the amount of connectivity and the appropriateness of those connections must both be tested**. [24]

In these studies, computation selects candidates for biological experiments. Genetic perturbation, cell targeting, connection tracing and functional tests then establish what those candidates mean. The next questions are whether an AI-identified pattern yields the same response to an intervention across laboratories, species and injury conditions, and whether an intervention feasible in people produces a lasting benefit.

9. How Jülich's maps and models help read an injured circuit

Knowing a cell's output destination does not reveal the condition of the wider brain into which it must integrate. That condition differs between individuals. Even with damage in the same location, surviving pathways and activity in distant regions may vary. Forschungszentrum Jülich brings anatomical mapping, imaging analysis and large-scale computation to this problem.

Julich-Brain organizes human brain regions, distinguished by the arrangement of their cells, into probabilistic maps. These provide a common anatomical reference while preserving variation between individuals. Jülich's INM-1 imaging AI, EBRAINS, and the modeling and computational support of the Jülich Supercomputing Centre complement this foundation. INM-1 is also developing models for microstructural imaging using JUPITER. Such infrastructure helps connect tissue and circuit data, but a map cannot reveal the history of new-neuron formation in a particular patient. [25,26]

A more direct connection to recovery comes from a 2021 study by Anna Bonkhoff and colleagues involving Jülich's INM-3. The team analyzed MRI data from 54 people with acute ischemic stroke; the six-month motor-recovery analysis included the 30 with available follow-up. A random-forest model used initial motor impairment together with information about how brain connectivity occupied and switched between states over time. For classification into three recovery levels, the internally cross-validated AUC was 0.84 with initial clinical scores alone and 0.89 when dynamic connectivity information was added. AUC measures how well a classifier discriminates between categories; it does not mean that 89% of patients recovered. [27]

The study suggests that the operation of surviving circuits can add prognostic information. External validation in independent hospitals and patient populations remains necessary, and the researchers did not test whether rehabilitation selected by AI was more effective. Predicting an observed course of recovery and identifying which treatment would change that course require different evidence. [27]

Mechanistic computational models have another role. In 2016, collaborators in Marseille, Toronto and UC Irvine built individualized models with The Virtual Brain using imaging from 20 people with chronic stroke and 11 controls. Equations represented connections between regions and the dynamics of neuronal populations. The researchers fitted model parameters to imaging signals and examined their relationship with subsequent changes in motor scores. The study neither counted new neurons directly nor clinically validated a treatment; its biophysical simulation is not itself a machine-learning AI model. It is a separate research example relevant to Jülich's present infrastructure, illustrating how models can generate hypotheses about the wider circuit effects of local cellular changes. [28]

The current Virtual Brain Twin project, in which Jülich participates, is likewise developing computational architecture and operating infrastructure for individualized models. Its stated goal is an ecosystem for personalized treatment of psychiatric disorders. That goal cannot be presented as a clinical achievement in brain-injury regeneration. The imaging analysis, random forests and virtual-brain simulations discussed here use classical computation. Running a large supercomputer does not imply that a quantum processor performed the calculation. [26]

10. Quantum technology first enters through observing recovery

Among quantum technologies, magnetic sensors provide the most concrete connection to this question. An optically pumped magnetometer, or OPM, uses light to prepare atomic spins and reads the optical response as the magnetic field changes. OPM-based magnetoencephalography, or OPM-MEG, measures the tiny magnetic fields generated by currents in neuronal populations. Its physical role differs from that of a quantum processor calculating a model of the brain. [29]

A 2025 study involving UCL used wearable OPM-MEG in three healthy men to detect changes in sensorimotor cortical beta activity at 15-30 Hz during a stepping task of approximately 0.5 m. It advances the ability to read circuit activity while the head moves. The result suggests a way to measure brain signals alongside physical actions during rehabilitation, although magnetic shielding and the suppression of movement-related noise remain necessary. The study did not test a treatment in patients, and beta-activity changes cannot be converted into a count of new neurons or an amount of regeneration. [29]

Patient research using brain signals as feedback has a longer history. In a 2008 study involving NIH, Tübingen and collaborators, conventional MEG based on superconducting quantum interference devices, or SQUIDs, recorded signals from eight people with chronic severe stroke to control a hand orthosis. Six learned to control the device, but clinical hand-function scales did not improve significantly. This was a different sensor system from today's wearable OPM arrays. Successful control of a device and improved function without that device after treatment are outcomes that each require assessment. [30]

Quantum computing introduces further questions about validation. CLARA, a research program involving the Paris Brain Institute, Leibniz Supercomputing Centre and other partners, aims to combine AI and quantum computing in brain research. It plans hybrid quantum-classical calculations of protein dynamics and molecular properties, with APOE, amyloid-β and tau in Alzheimer's disease among its main targets. It has not presented clinical evidence of improved new-neuron integration or rehabilitation after brain injury. Jülich's classical neuroscience computation and the plans of this separate program must each be evaluated at their own stage of implementation. [31]

For a quantum processing unit, or QPU, to become a practical tool here, researchers must specify the molecular or circuit calculation it performs and compare its accuracy and total cost with the best classical methods under matched conditions. Any computational benefit must then be tested for its contribution to a biological intervention and functional recovery. Quantum sensors already have a concrete experimental role in measuring activity. A regenerative-treatment benefit from QPU computation remains a separate research question.

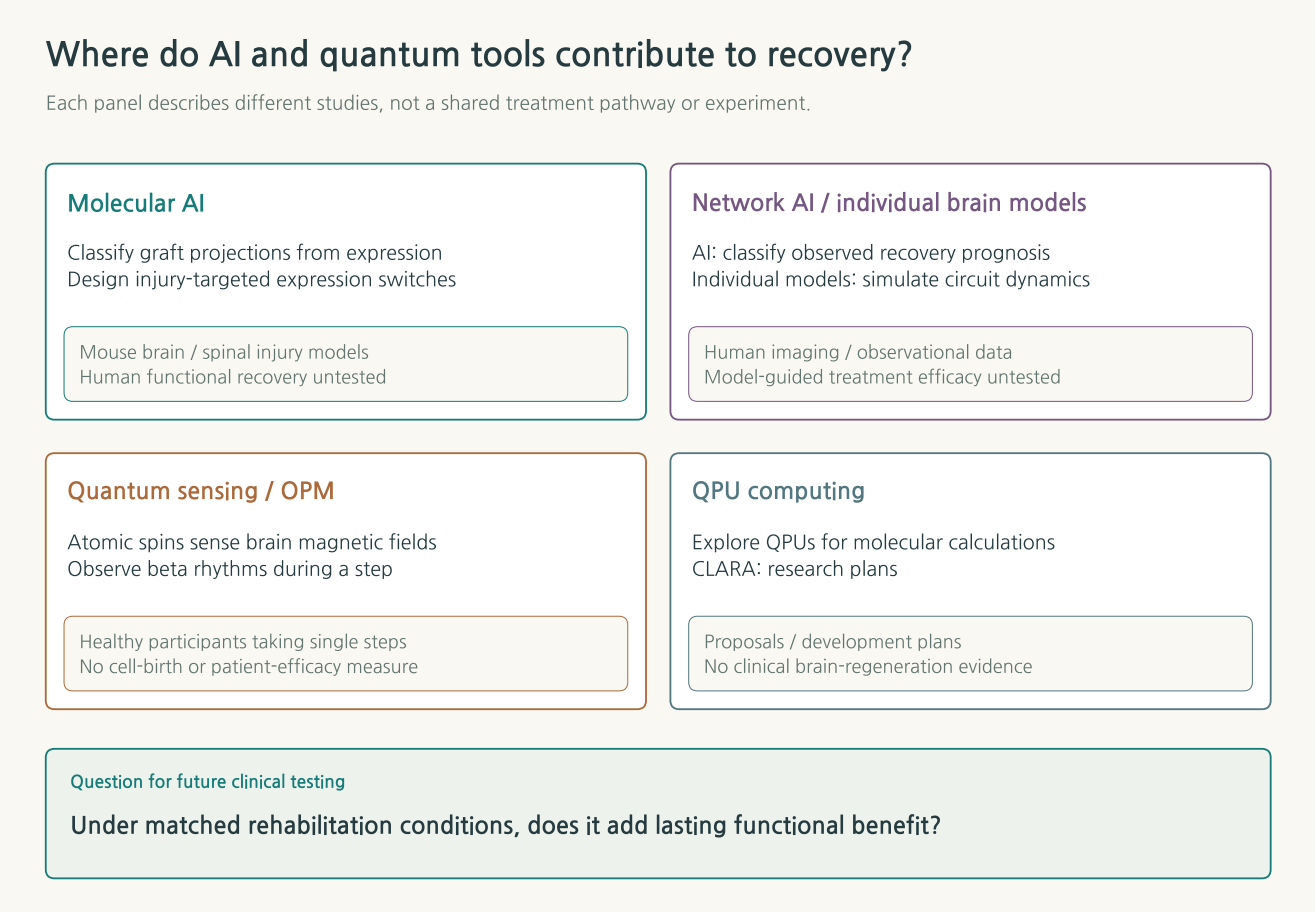


Figure 4. The roles of AI and quantum technologies are compared through the same question about recovery. Mouse experiments on cellular targeting and projection patterns, patient prognostic models, activity measurements in healthy participants, and computational research plans occupy different stages of validation. The diagram does not represent a study linking all these stages or rank their treatment effects. Explanatory diagram constructed for this review from primary sources and official research plans. [21,23,27,28,29,31]

In Figure 4, molecular AI asks what to change; network analysis asks which circuits remain; and sensors ask how those circuits operate during a task. Building these kinds of evidence together can make experiments connecting regeneration and rehabilitation more specific. Their outputs cannot be placed on a common “recovery rate” scale. The next experiment must establish what each result contributes to functional recovery.

11. The next advance will connect cell birth with cell use

A first challenge is to measure neurogenesis specifically enough in the living human brain. Blood markers, imaging, and molecular analyses offer possibilities for repeated observation, but researchers must establish which parts of birth, survival, and maturation each measurement reflects. Linking molecular states found in postmortem tissue to a living person's course could help identify when and for whom rehabilitation or a treatment works.

A second challenge concerns the quality of functional integration. Animal studies can label new cells, measure their electrical properties alongside behavior, and selectively manipulate their activity to test a causal contribution. Direct observations are more restricted in humans, so tissue, imaging, physiology, and clinical outcomes must complement one another. Connecting cell numbers, axonal direction, excitation and inhibition,

and actual learning could identify more useful treatment conditions than simply stimulating proliferation. Targets found by molecular AI, predictions from brain models and population activity read by magnetic sensors concern different scales. Comparing cells, circuits and behavior before and after the same intervention is needed to validate the relationships between them.

A third challenge is to establish what a biological intervention adds to rehabilitation. Future trials of cell therapy or methods intended to promote regeneration need to record the rehabilitation received by both groups, its content and amount, time since injury, and baseline function. Alongside tests of efficacy, evaluating everyday tasks and participation, durability, fatigue, and adverse effects can reveal what changes persist in a person's life. Combining new cells with learning experience is a promising research direction, but any advantage of that combination requires comparison under matched conditions. One approach would be a prospective comparison between rehabilitation guided by a model and rehabilitation without that guidance. This would test whether accurate prognosis also helps select a useful treatment.

Taken together, the available evidence suggests that near-term progress is more likely to come from **better matching rehabilitation that uses surviving circuits with treatments directed at specific biological bottlenecks** than from a single technology that rebuilds the entire brain. This is the review's outlook. Independent clinical findings showing that interventions preserve cells, support appropriate connections, and add lasting functional benefit to usual rehabilitation could turn that outlook into more specific treatment strategies.

Carbon in the atmosphere revealed when cells were born. Understanding the time it takes a brain to learn again requires asking who those cells connect with, what they do, and which changes in everyday life they make possible. Adult neurogenesis has given science a tangible subject for that longer investigation. Rehabilitation works from its other end, establishing, one task at a time, what a person can do again.

This scientific review explains public research and clinical evidence. It does not assess an individual's prognosis or prescribe treatment.

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How this review was made

Responsible editor	Hyun-Jung Kim
AI system	OpenAI Codex; exact model identifier not retained
Verifiable agent roles	root — primary-source synthesis, Korean writing, original conceptual cover, AI/quantum extension and publication integration; academic_sources — human hippocampal evidence and denominators, Jülich/personalized-brain-model sources and independent scientific audit; rehab_evidence — rehabilitation/cell-therapy clinical audit, AI cellular/axon-repair sources and complete English/extension translation; review_format — original bilingual responsive figures, font typesetting, HTML/PDF production and complete output checks; publish_audit — quantum-sensing/QPU evidence, independent cover, reader-explanation, APS-reference and publication audit
Editorial harness	AI Tech Review Editorial Harness v2026.08 · public method
Inputs and outputs	Public papers, regulatory reviews, rehabilitation guidance and official disclosures → bilingual review, conceptual cover, explanatory figures and PDFs.
Production request	Review adult human neurogenesis, recovery and rehabilitation, and extend the same cell/circuit/function question to actual AI and quantum-technology approaches at major research institutions.
Verification scope	Accessible sources, versions, samples, methods and reported values; New Scientist was checked through its public introduction, the killifish study through its public abstract, and some Moreno-Jiménez evidence through the abstract and extended figures; original bilingual figures and HTML/PDF output. The extension checked original AI graft, injury-responsive enhancer and axon-repair studies, Jülich maps/prognostic models, OPM/SQUID reports and official QPU research plans. Wang was checked through the final abstract, author-deposit descriptions and public preprint v2 of 28 October 2025; the final full text and notebook contents were not accessed. No independent tissue/single-cell raw-data analysis, rerun of the studies' computational code, clinical trial or rehabilitation treatment.
Human review	Topic, writing and public posting requested; no line-by-line human approval or reader-comprehension study.
Evidence cutoff	2026-10-30

Cell-birth observations, circuit-function studies and clinical endpoints are distinguished. The AI-generated cover is conceptual, not anatomical imaging, cell imaging or a treatment result.